Question:

Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

<think> To solve this problem, let's first calculate the new value of the house after Josh made the repairs. According to the problem, the repairs increased the value of the house by 150%. Since Josh bought the house for $80,000 and then added $50,000 in repairs, we need to first determine the original value of the house before the repairs were made. However, we don't need to calculate the original value separately since we are given that the repairs increased its value by 150%.

Let's denote the original value of the house as V. After the repairs, the new value of the house is $V + 1.5V = 2.5V$. We know that after the repairs, the house was worth $80,000 + 50,000 = 130,000$. So, we can set up the equation $2.5V = 130,000$ to find V. However, we can also directly calculate the profit by comparing the final value of the house to the total amount Josh spent to buy and repair it.

Josh's total expenditure includes the original cost of the house and the cost of repairs. So, we add the cost of the house and the cost of repairs together: $80,000 + 50,000 = 130,000$. Now, since we already know the final value of the house is $130,000, and this final value represents the original cost plus the 150% increase, we can see that the profit is simply the final value minus the total initial investment.

Therefore, to find the profit, we subtract the total initial investment from the final value: $130,000 - 130,000 = 0$. However, this approach contradicts the logic we need to follow. Let's reconsider by understanding that the 150% increase means the house's value became 2.5 times its original value plus the cost. But since the problem asks for profit, we directly compare the final value with the original cost:

$$\text{Profit} = \text{Final Value} - \text{Total Initial Investment}$$
$$\text{Final Value} = 130,000$$
$$\text{Total Initial Investment} = 80,000 + 50,000 = 130,000$$

So, the profit calculation would be:
$$\text{Profit} = 130,000 - 130,000 = 0$$

However, this logic again doesn't make sense because it implies no profit which seems incorrect based on the context. Let's recalculate the final value correctly by considering the whole increase of 150% on the original value which wasn't clarified before. We previously stated $2.5V = 130,000$, but given the 150% increase on the original cost (80,000), it simplifies to:
$$V = \frac{130,000}{2.5} = 52,000$$
Since the original cost was $80,000$ plus the repair cost $50,000$, the total cost was $80,000 + 50,000 = 130,000$.

Given the direct comparison, we should revise the profit calculation by focusing on how the entire increase is calculated correctly without using the V intermediary.

Thus, since the increase is directly given as 150% of the original cost plus